



PROBLEM STATEMENT

To analyze Spotify song data and identify patterns in audio features, popularity, and trends — uncovering what musical characteristics make songs popular across years and languages.



Types of Data Analysis in Spotify Dataset

Univariate Analysis:

Examines individual song features (like danceability, energy, valence) to understand their overall distribution and trends.

Bivariate Analysis:

Explores relationships between two variables — e.g., how energy relates to popularity or tempo affects danceability.

Multivariate Analysis:

Studies the combined effect of multiple audio features to identify clusters or patterns in popular songs.

Time Series Analysis:

Analyzes how average song properties (like loudness, tempo, and duration) have evolved over the years.

Statistical Analysis of Variables

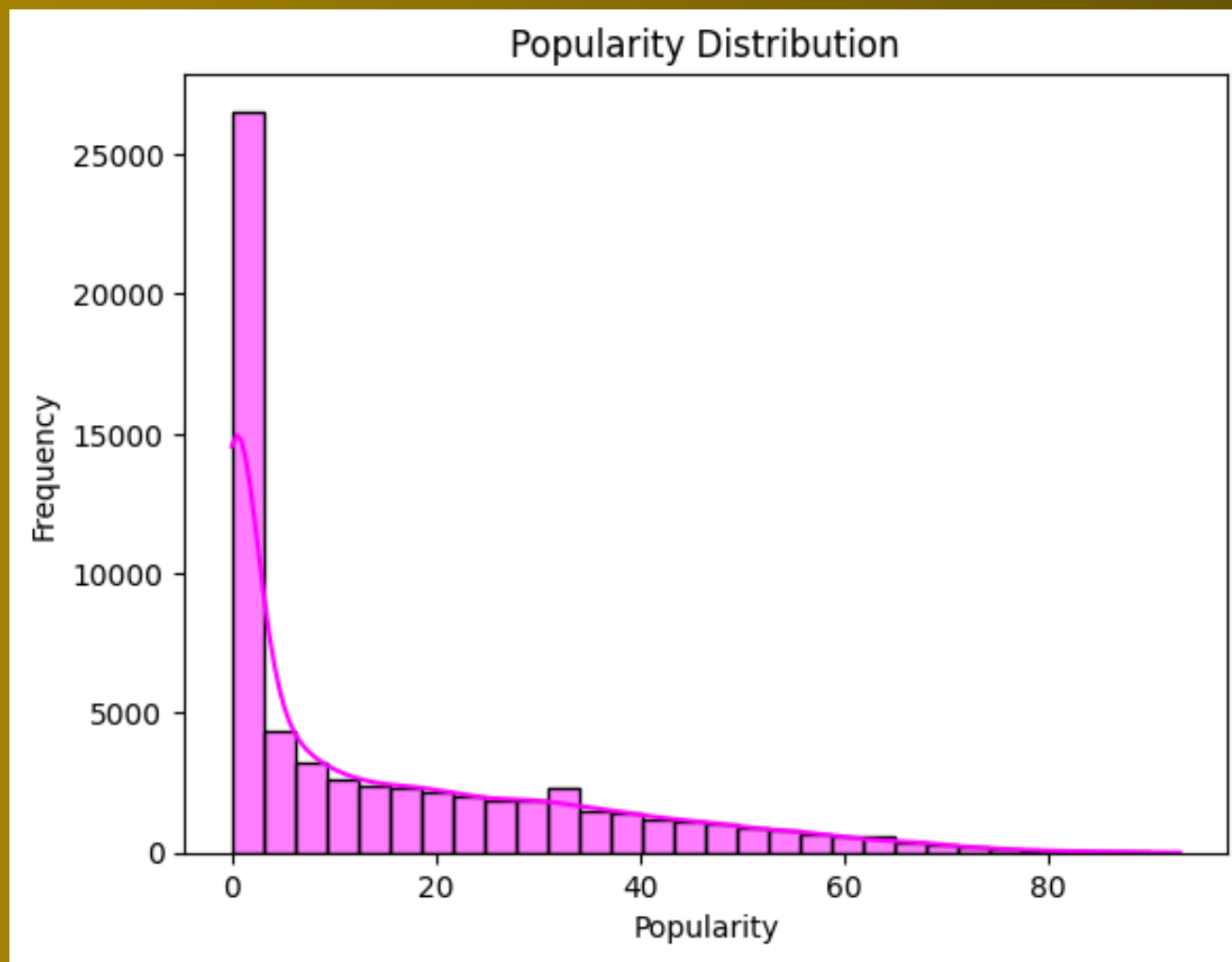
index	year	popularity	acousticness	danceability	duration_ms	energy	instrumentalness
count	62317.0	62317.0	62317.0	62317.0	62317.0	62317.0	62317.0
mean	2014.4259351380842	15.358361281833208	0.3622916802588379	0.5968067044305727	242527.040021182	0.602496405887639	0.14621451272269206
std	9.645112828953911	18.62690786061795	0.3146093436386704	0.18620863095267362	112999.9326066876	0.24614448781727766	0.30780356426048766
min	1971.0	0.0	-1.0	-1.0	5000.0	-1.0	-1.0
25%	2011.0	0.0	0.0671	0.497	192160.0	0.44	0.0
50%	2017.0	7.0	0.286	0.631	236267.0	0.639	2.54e-05
75%	2022.0	26.0	0.632	0.73	286240.0	0.803	0.0152
max	2024.0	93.0	0.996	0.986	4581483.0	1.0	0.999

index	key	liveness	loudness	mode	speechiness	tempo	time_signature
count	62317.0	62317.0	62317.0	62317.0	62317.0	62317.0	62317.0
mean	5.101657653609769	0.1941425052553878	-8.972681724088131	0.5860519601392878	0.08772197313734616	117.93124681868512	3.8570855464800937
std	3.553469049692961	0.1720304390149833	5.464688446758133	0.4936823474200719	0.11514978329527352	28.509458913714433	0.5026600382113825
min	-1.0	-1.0	-60.0	-1.0	-1.0	-1.0	-1.0
25%	2.0	0.0932	-10.727	0.0	0.0367	95.942	4.0
50%	5.0	0.125	-7.506	1.0	0.0489	117.991	4.0
75%	8.0	0.243	-5.456	1.0	0.0891	135.081	4.0
max	11.0	0.998	1.233	1.0	0.959	239.97	5.0

index	valence	duration_min
count	62317.0	62317.0
mean	0.49522618868045637	4.042117333686367
std	0.2647869593370689	1.8833322101114454
min	-1.0	0.08333333333333333
25%	0.292	3.2026666666666666
50%	0.507	3.9377833333333334
75%	0.71	4.770666666666667
max	0.995	76.35805

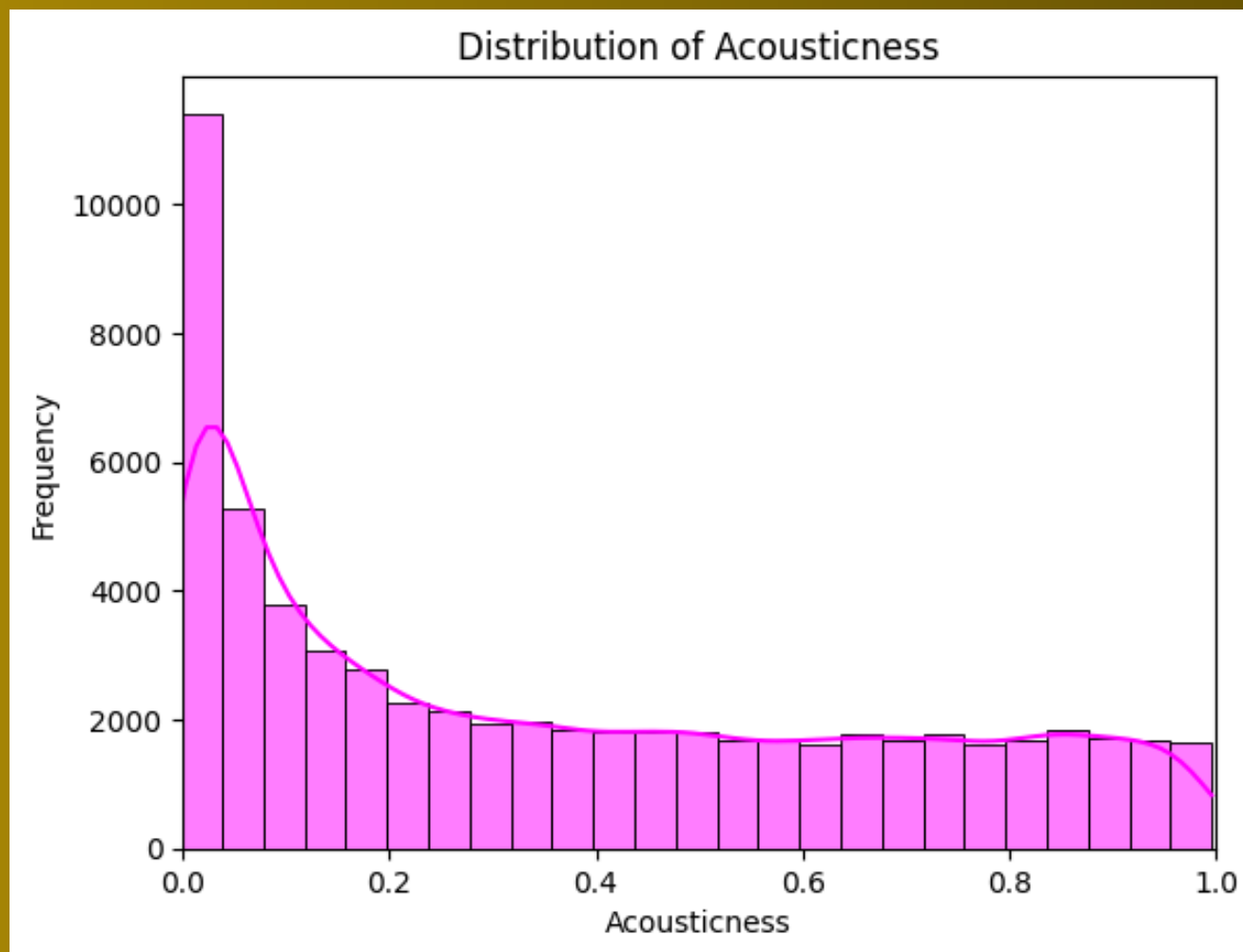
Popularity Distribution

The histogram shows that the vast majority of data points have a very low "Popularity" score, with a steep drop-off as the score increases. The distribution is highly right-skewed (or positively skewed).



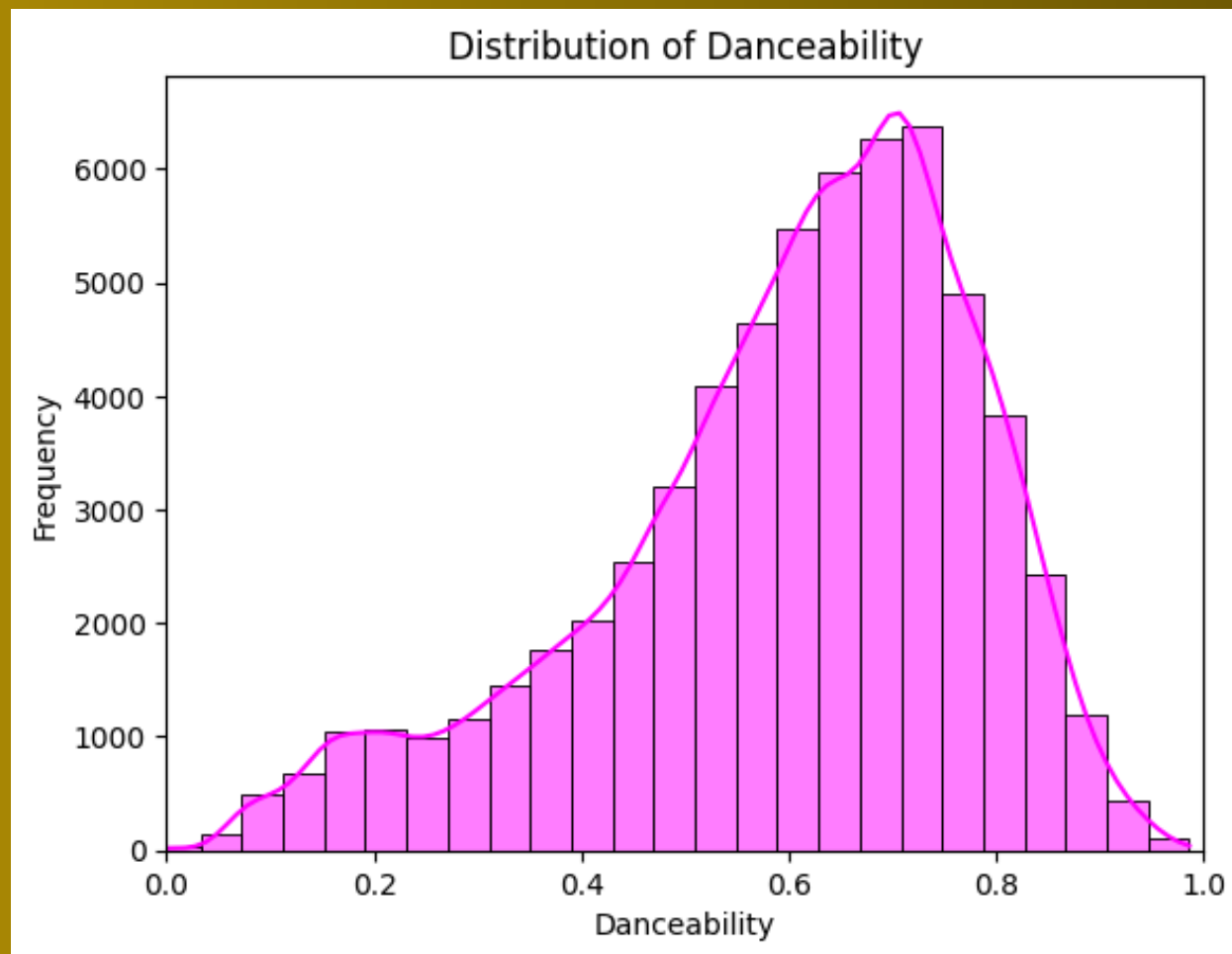
Acousticness Distribution

The distribution of "Acousticness" is bimodal, with the highest frequency being for values very close to zero (not acoustic) and a secondary, broader peak in the high acousticness range (around 0.8 to 1.0).



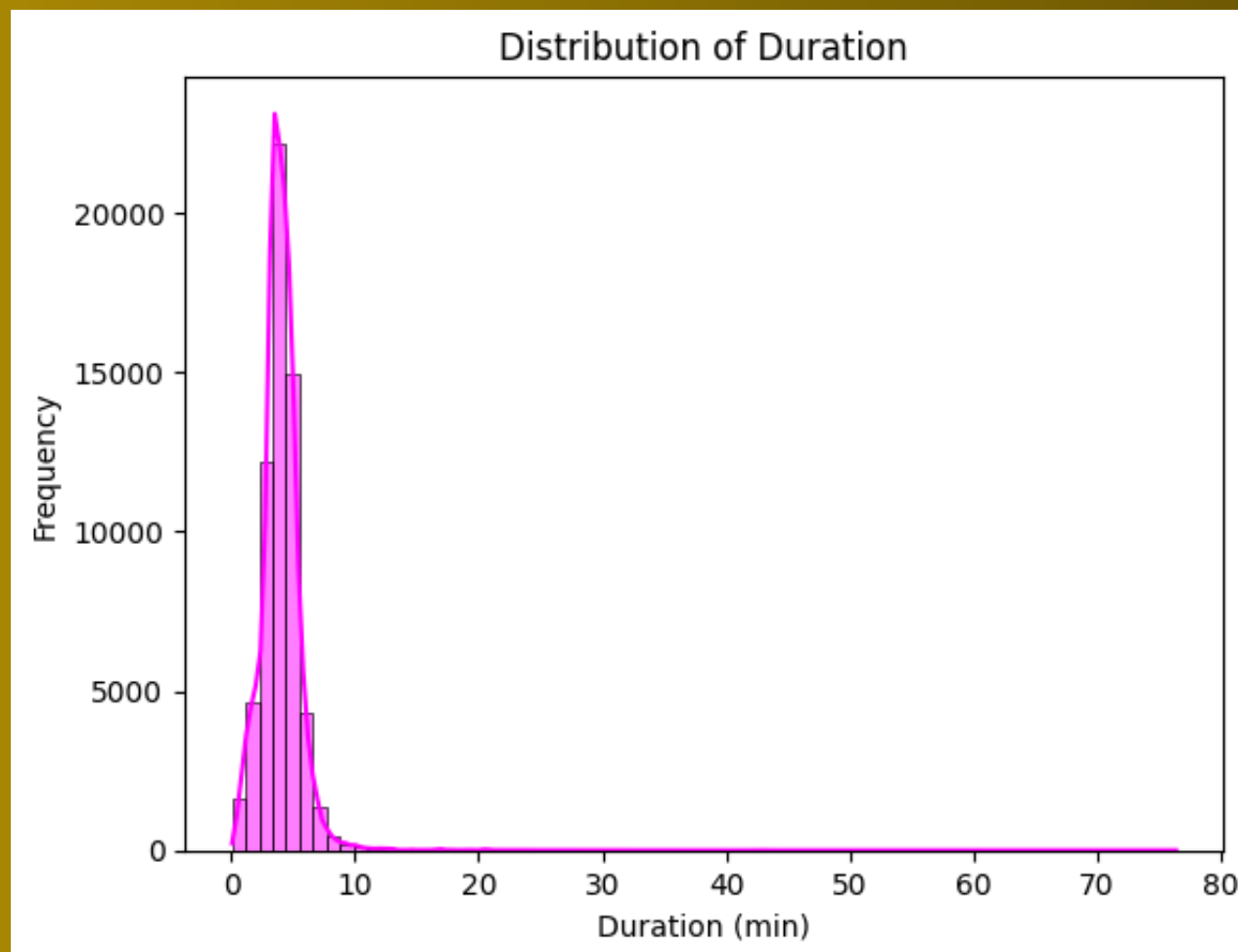
Danceability Distribution

The distribution of "Danceability" is left-skewed (or negatively skewed), indicating that most data points have moderately high to high danceability scores, with the peak frequency occurring around the 0.7 to 0.75 range.



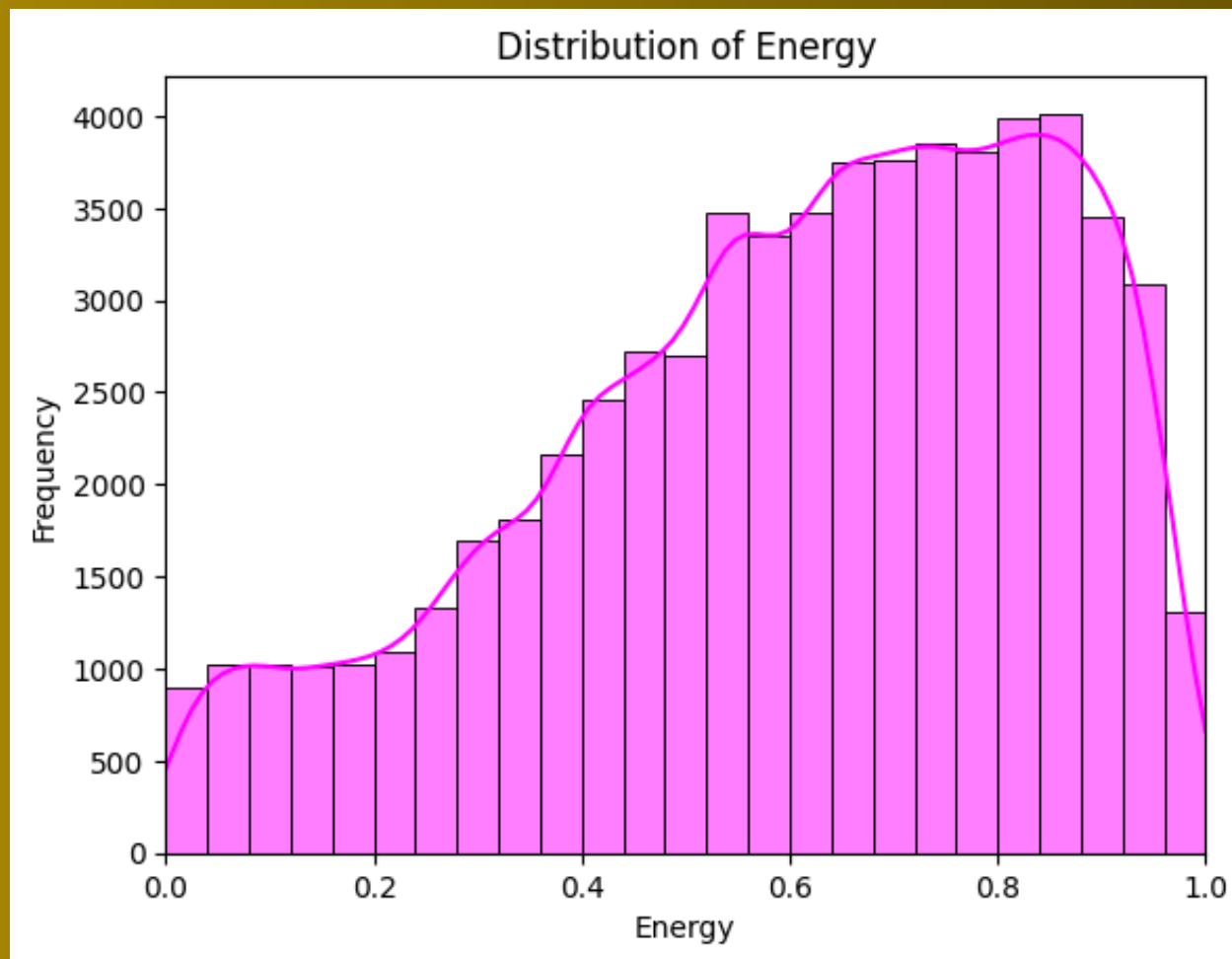
Song Duration Distribution

The distribution of "Duration" is extremely right-skewed (positively skewed), showing that the vast majority of durations are very short (under 10 minutes), but there are a few very long songs, including one that is approximately 76 minutes long.



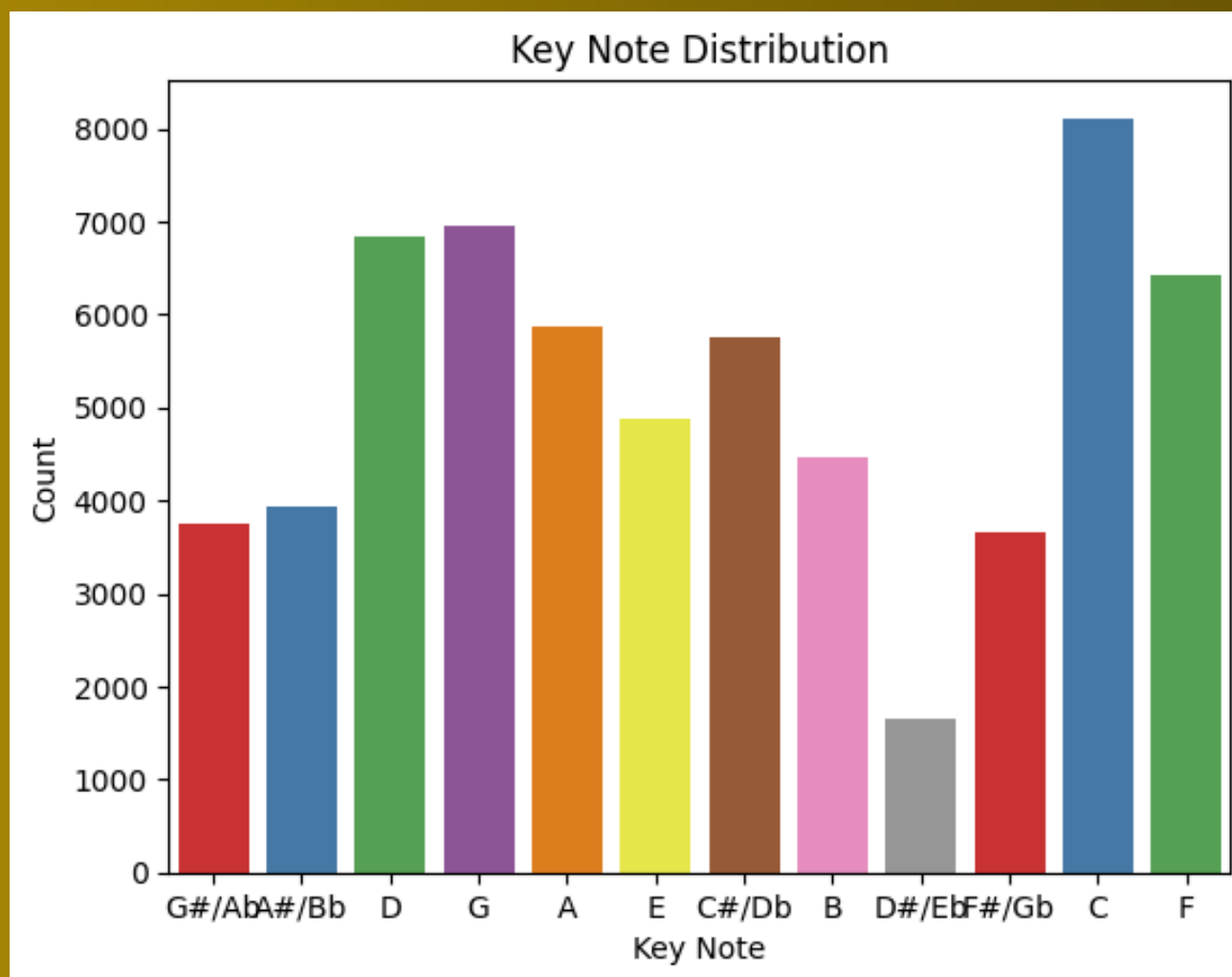
Energy Distribution

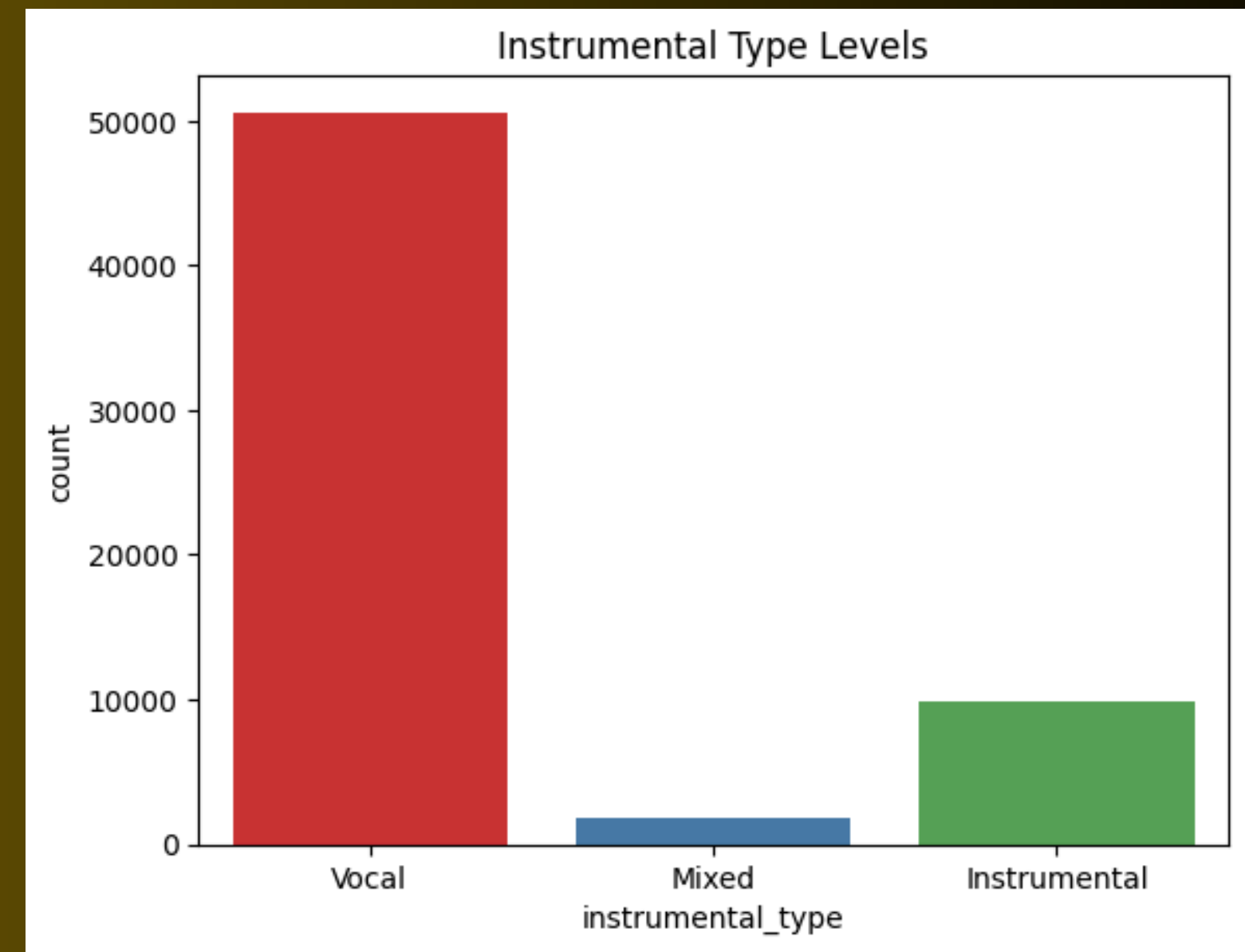
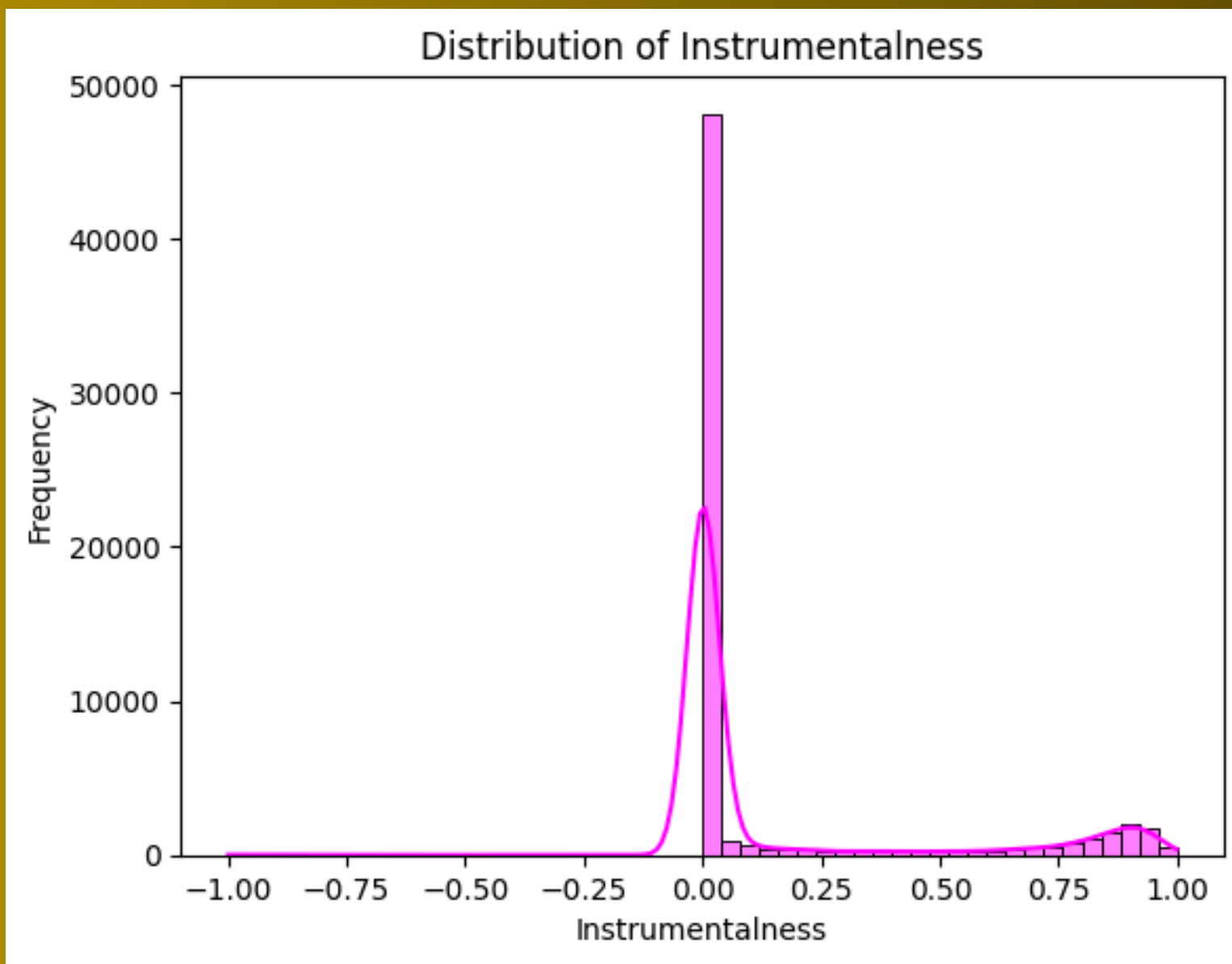
The distribution of "Energy" is strongly left-skewed (negatively skewed), indicating that most data points have high energy scores, with the frequency peaking around the 0.8 to 0.9 range.



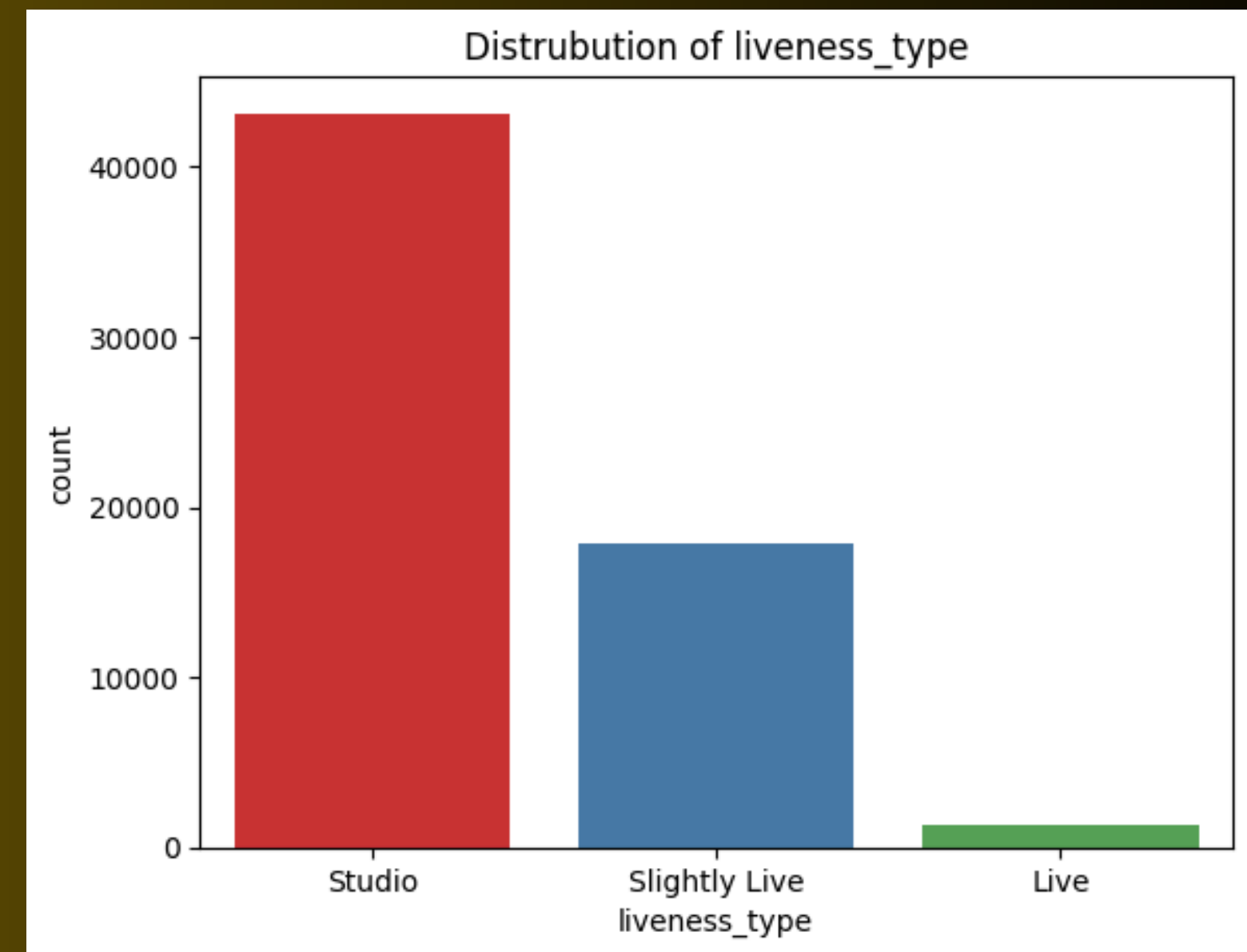
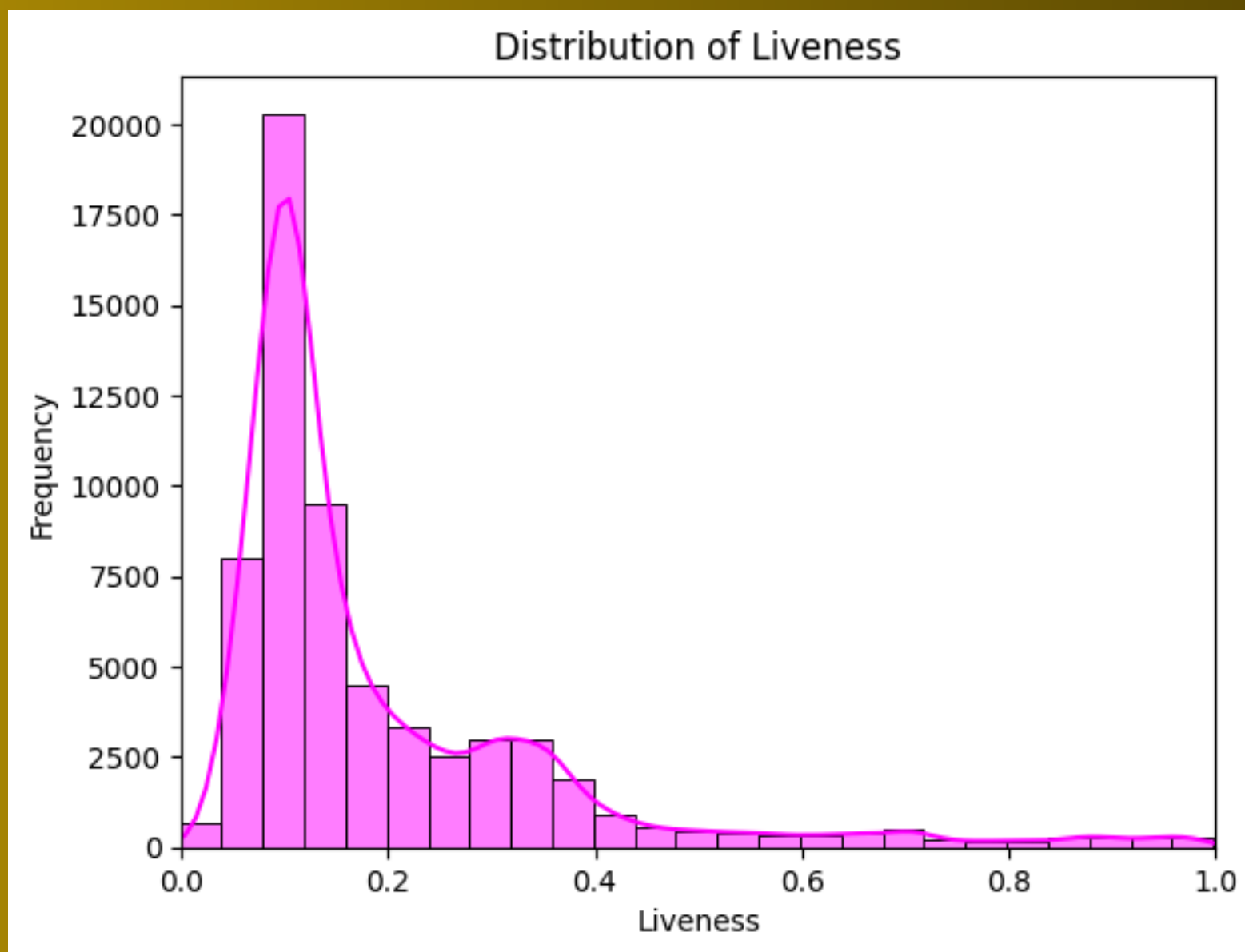
Key Note Distribution

The most frequent key note is C, with a count of over 8,000, followed closely by G and D (both around 7,000), indicating these are the most popular key signatures in the dataset.

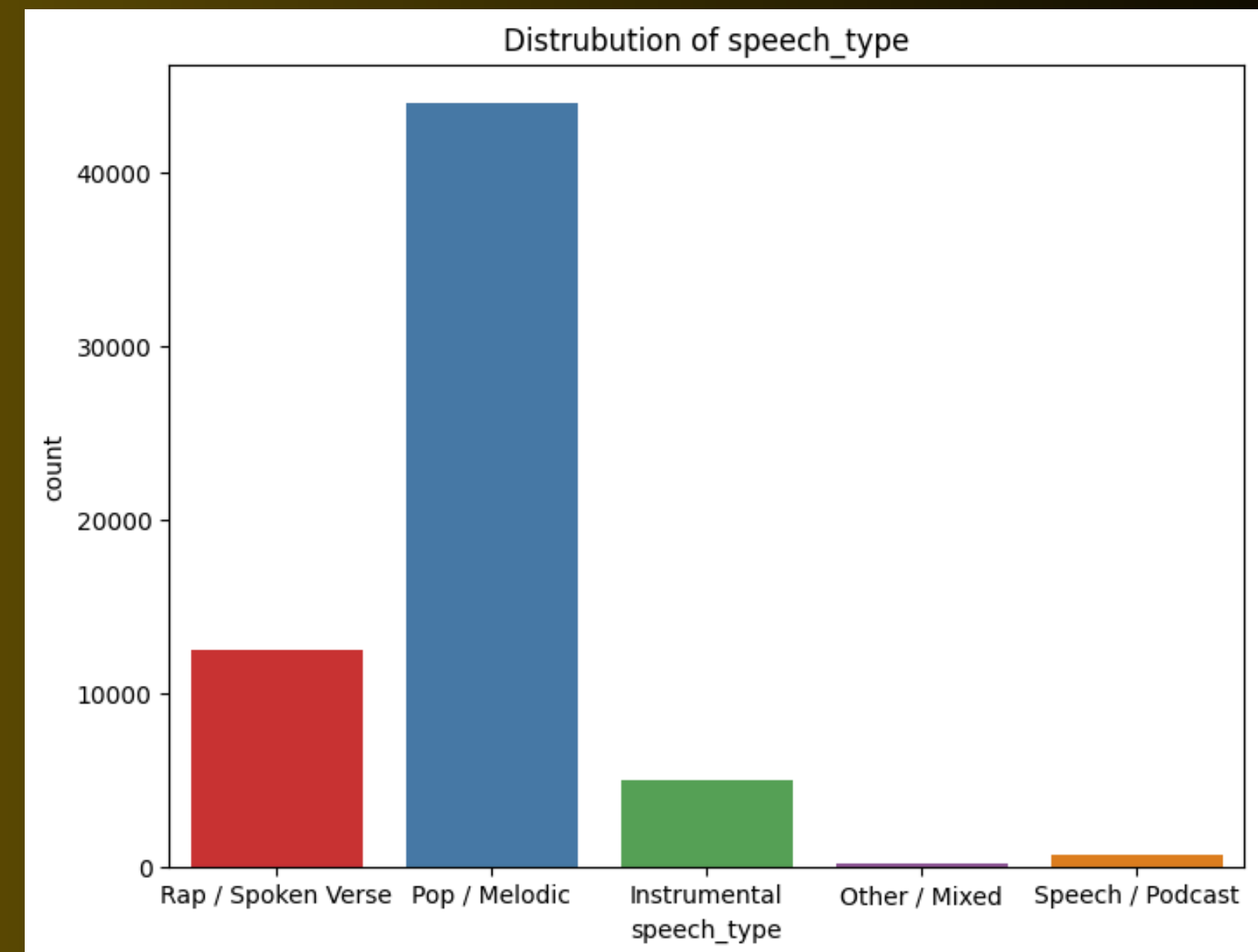
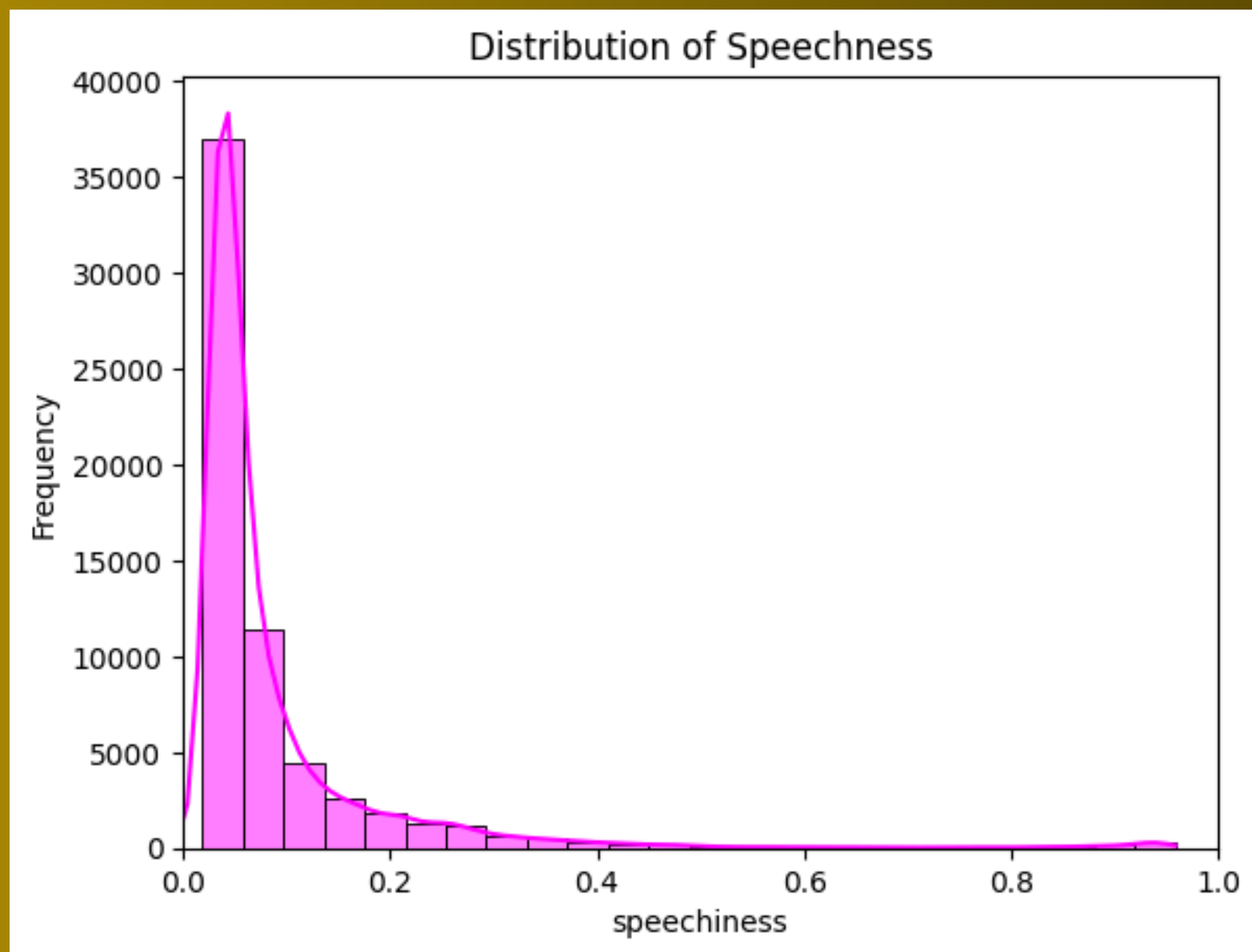




Both charts indicate a strong preference for vocal tracks, as the histogram shows a massive peak for an "Instrumentalness" score near zero (non-instrumental), and the bar chart confirms that Vocal tracks account for the overwhelming majority of the dataset, with a count of approximately 50,000.



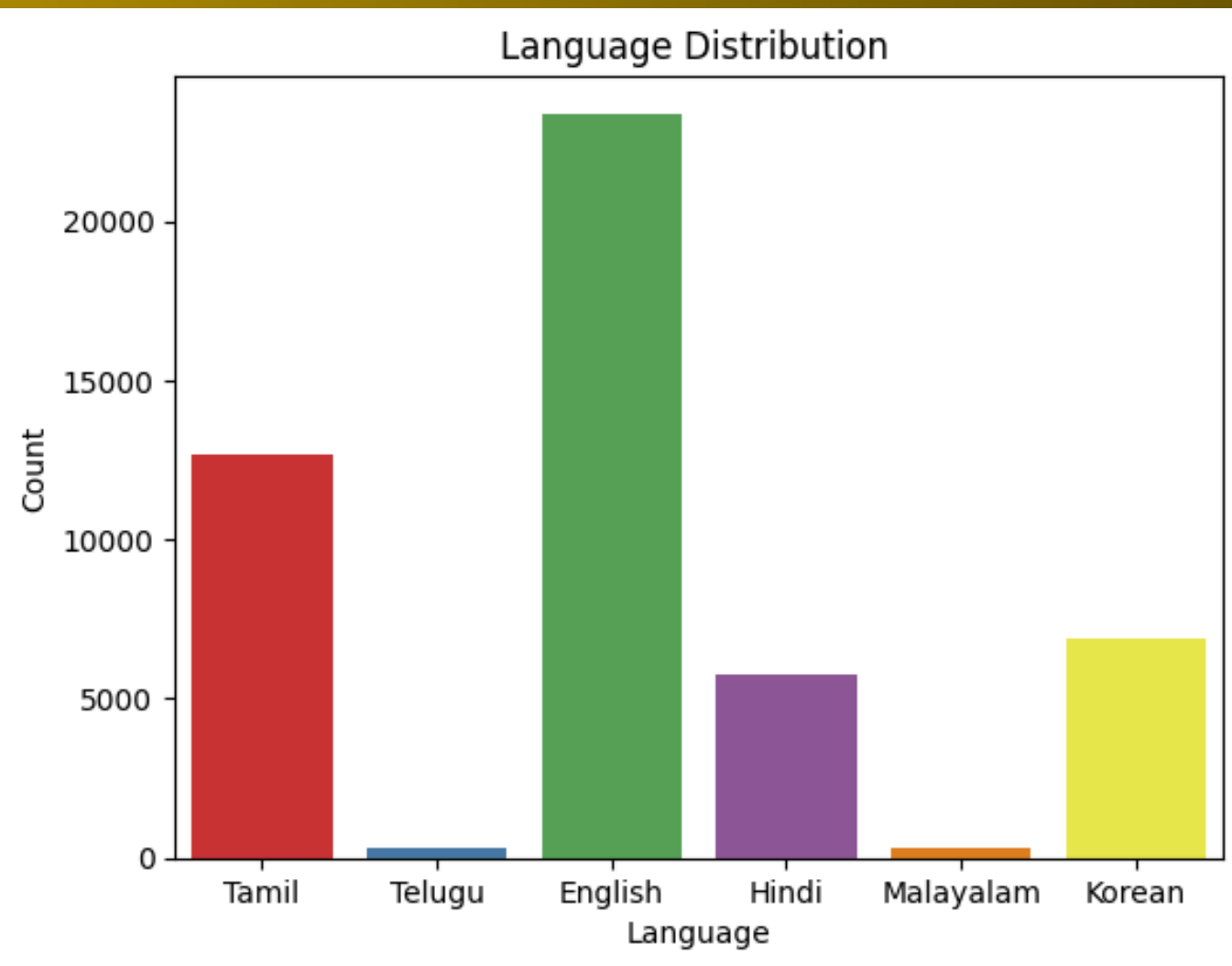
Both visualizations demonstrate that the majority of tracks are studio recordings, with the histogram showing an extremely high frequency peak for "Liveness" scores near zero and the bar chart confirming that Studio tracks have the highest count (over 40,000).



Both charts show that the vast majority of tracks are non-speech dominant, with the histogram peaking sharply near zero "Speechiness" and the bar chart confirming that Pop/Melodic tracks, which have the lowest speech content, dominate the dataset with a count over 40,000.

Language Distribution

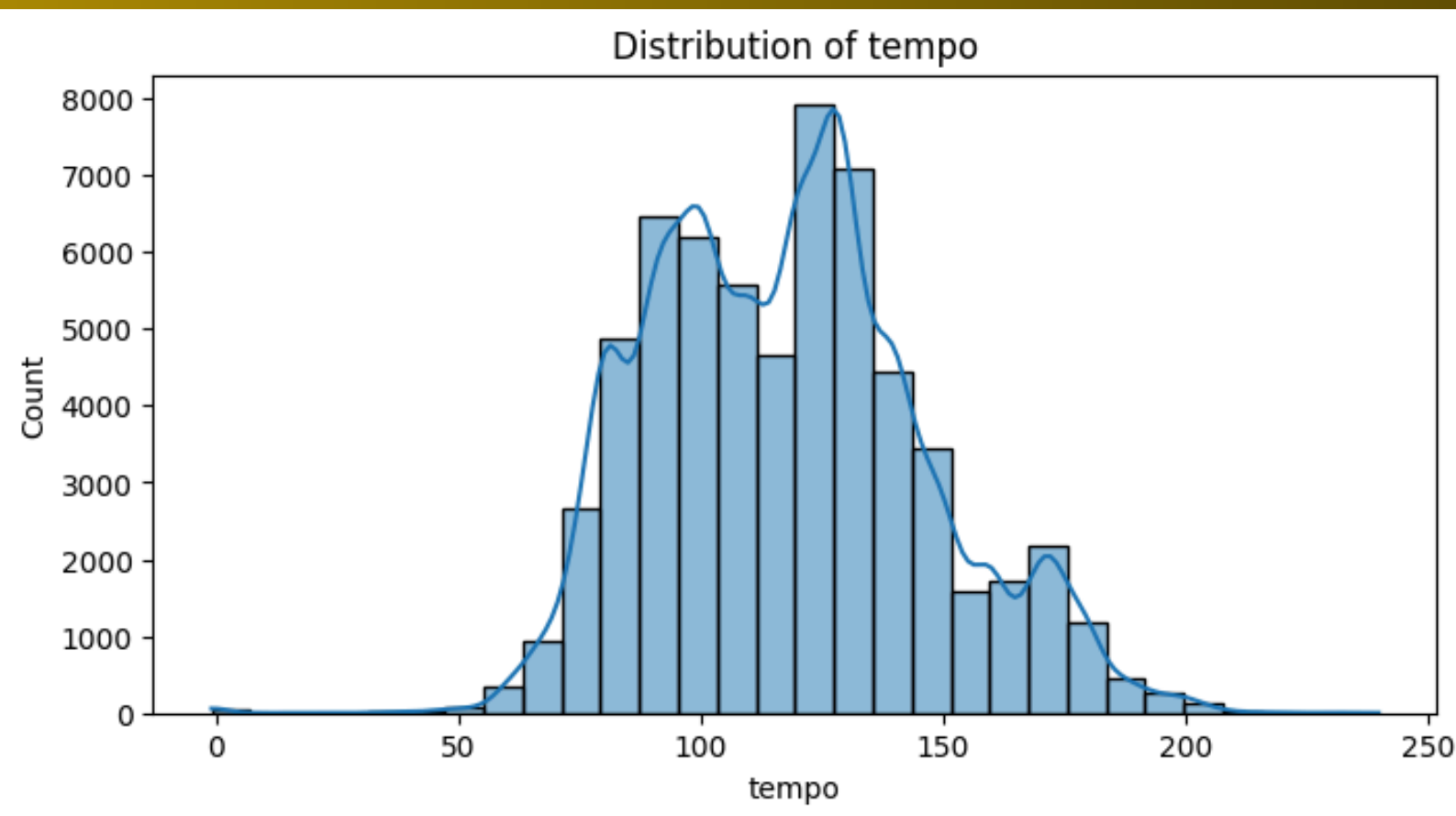
English has the highest count by a substantial margin, with a distribution of well over 20,000. The next highest language is Tamil, which is around 12,500.



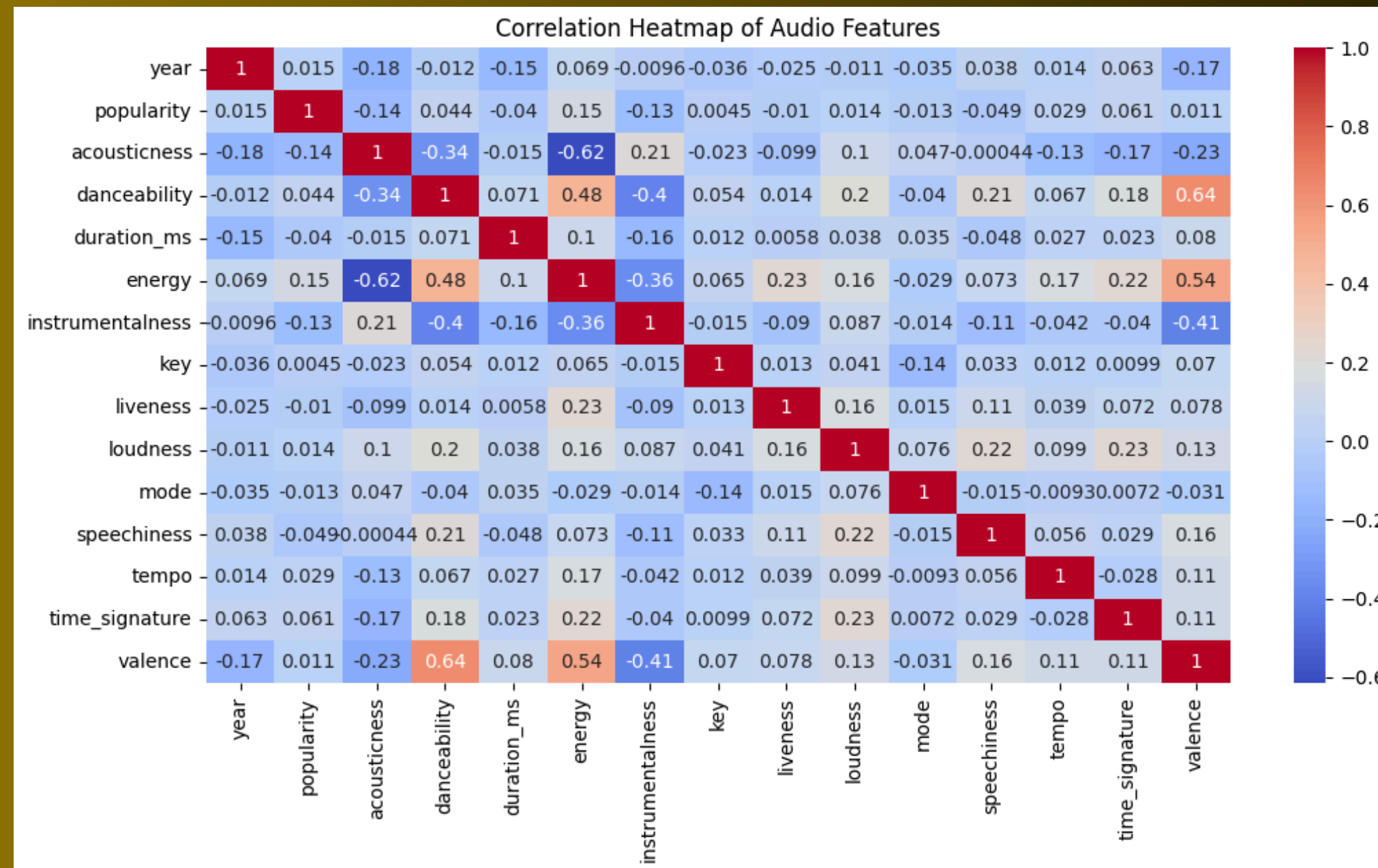
Tempo Distribution

The data shows two primary clusters or popular ranges for tempo:

- 1. A major peak centered around 120-130, with the highest count approaching 8,000.**
- 2. A secondary, slightly smaller peak centered around 95-105, with a count around 6,000-6,500.**



Correlation Heatmap Of Audio Features

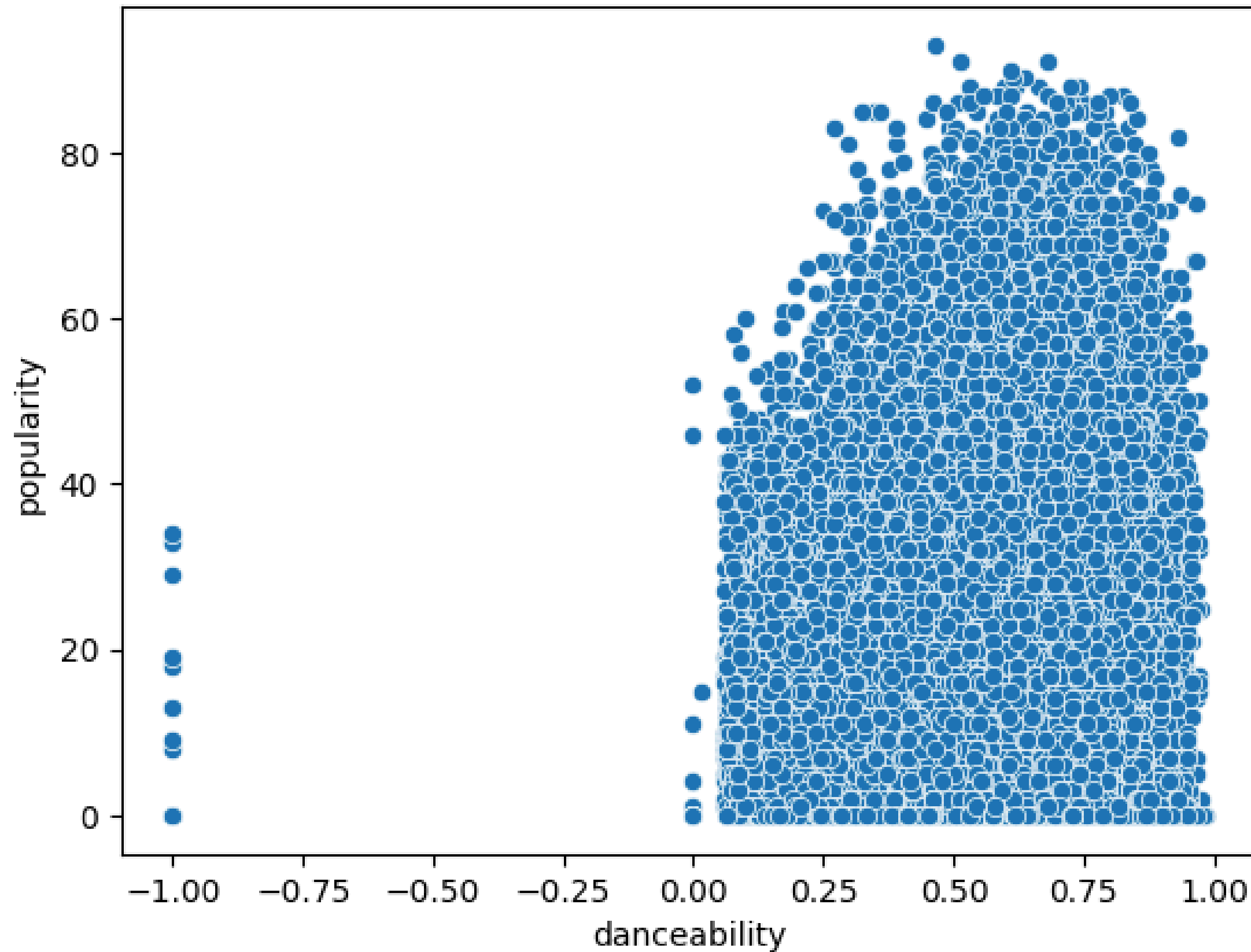


The overall insight from the Correlation Heatmap is that Energy, Loudness, and Acousticness are the most strongly related audio features.

Specifically:

1. Acousticness has a strong negative relationship with Energy (-0.62).
2. Energy has a strong positive relationship with Loudness (0.76).
3. This indicates that tracks that are less acoustically natural tend to be high in both energy and volume.

Popularity vs Danceability

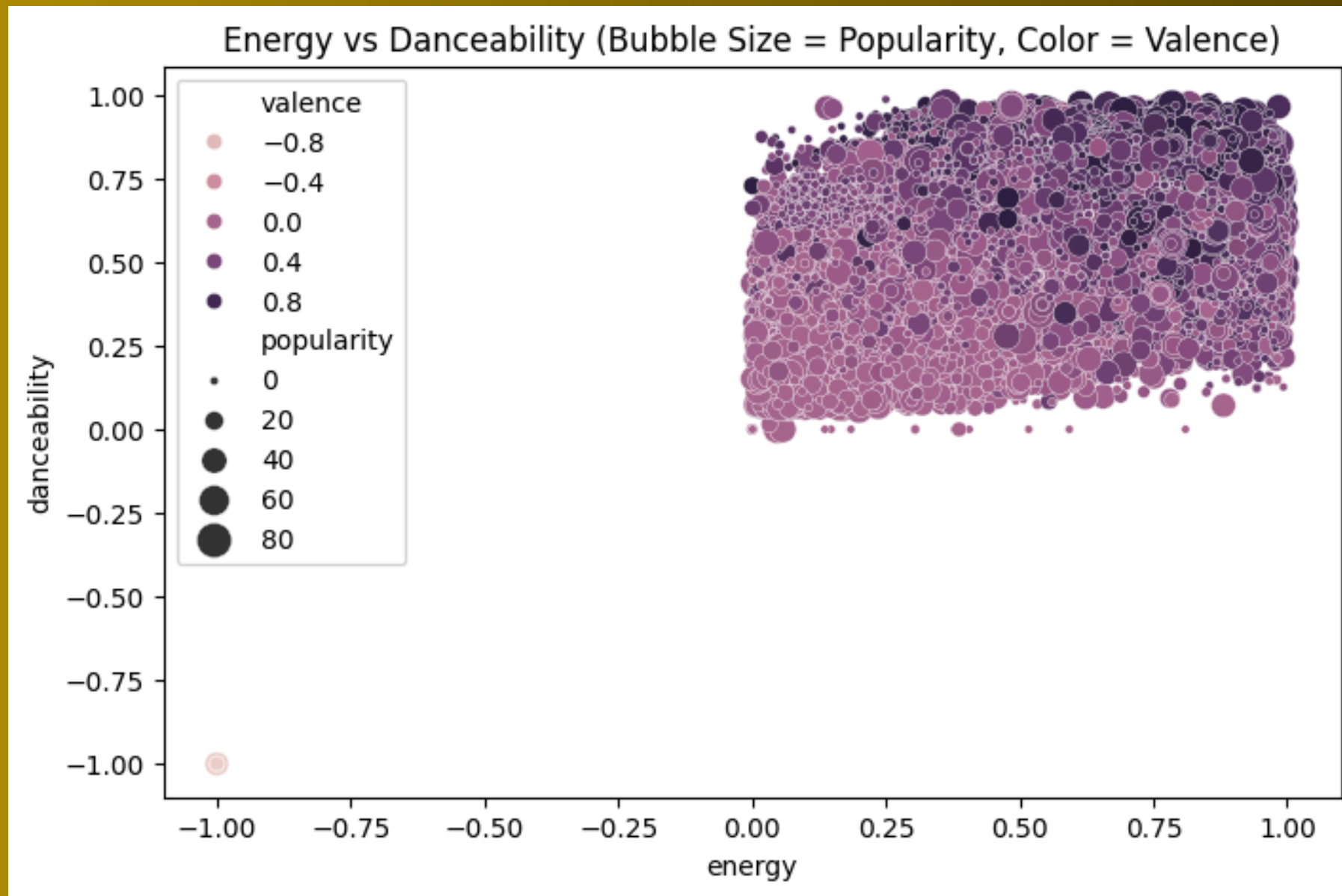


Based on the "Popularity vs Danceability" scatter plot, the main insight is that almost all songs in the dataset have a Danceability score of 0 or higher.

Furthermore, within this main cluster:

- Danceability scores are heavily concentrated between approximately 0.5 and 1.0.
- The highest Popularity scores (above 80) are mostly found in songs with a Danceability score between 0.5 and 0.9.

How Energy and Danceability Influence Popularity

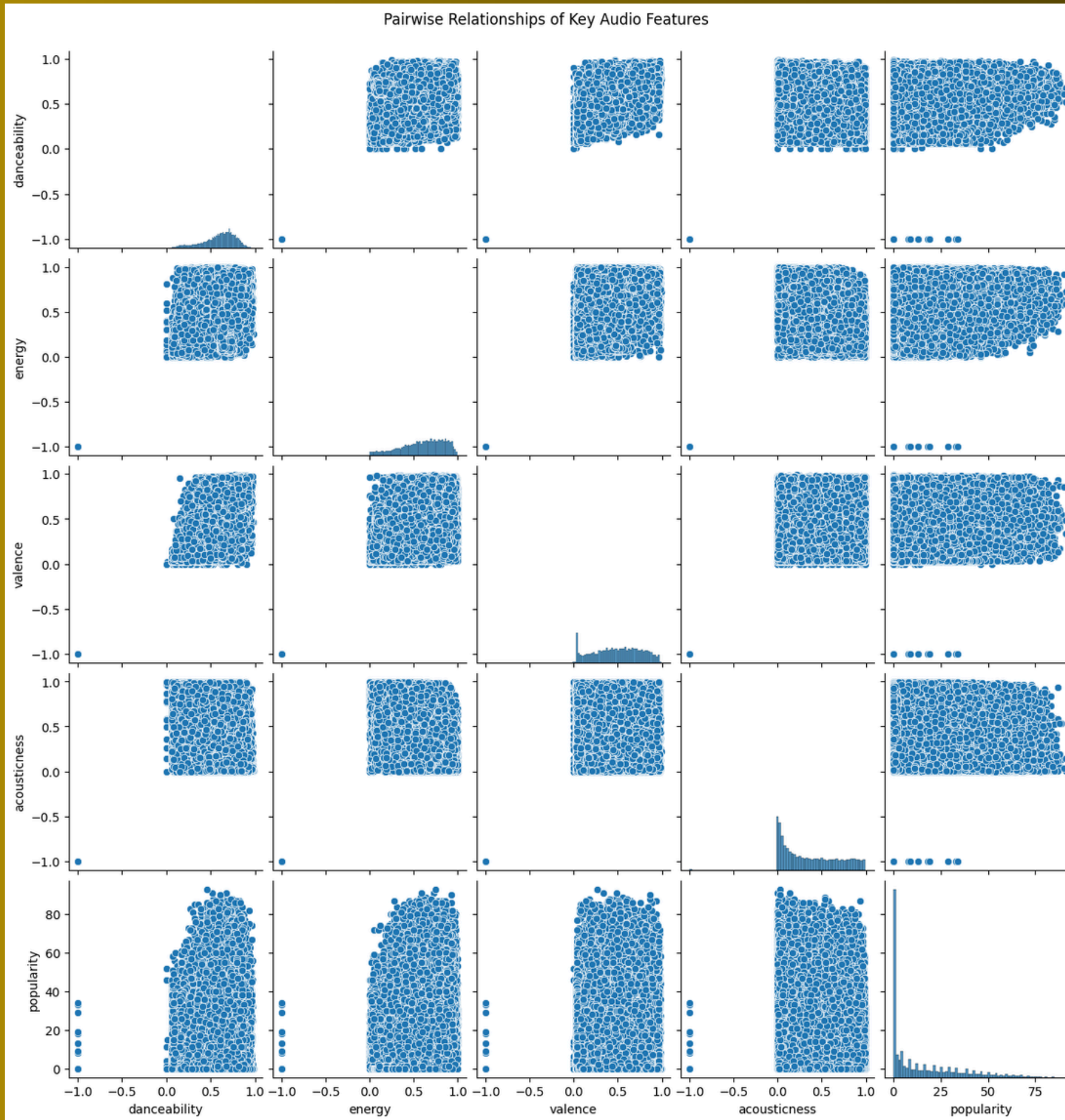


Based on the "Energy vs Danceability" bubble chart, the main insight is the strong, general co-occurrence of moderate-to-high Energy and moderate-to-high Danceability across the dataset.

Key observations from the main cluster (Energy ≥ 0 and Danceability ≥ 0):

- Most popular songs (largest bubbles) are found in the region of high Energy and high Danceability (top-right corner).
- Valence (color) appears to increase (darker purple, more positive mood) as both Energy and Danceability increase towards the top-right corner.
- The data is overwhelmingly concentrated where both Energy and Danceability are non-negative (greater than 0).

Pairwise Relationships Between Key Audio Features

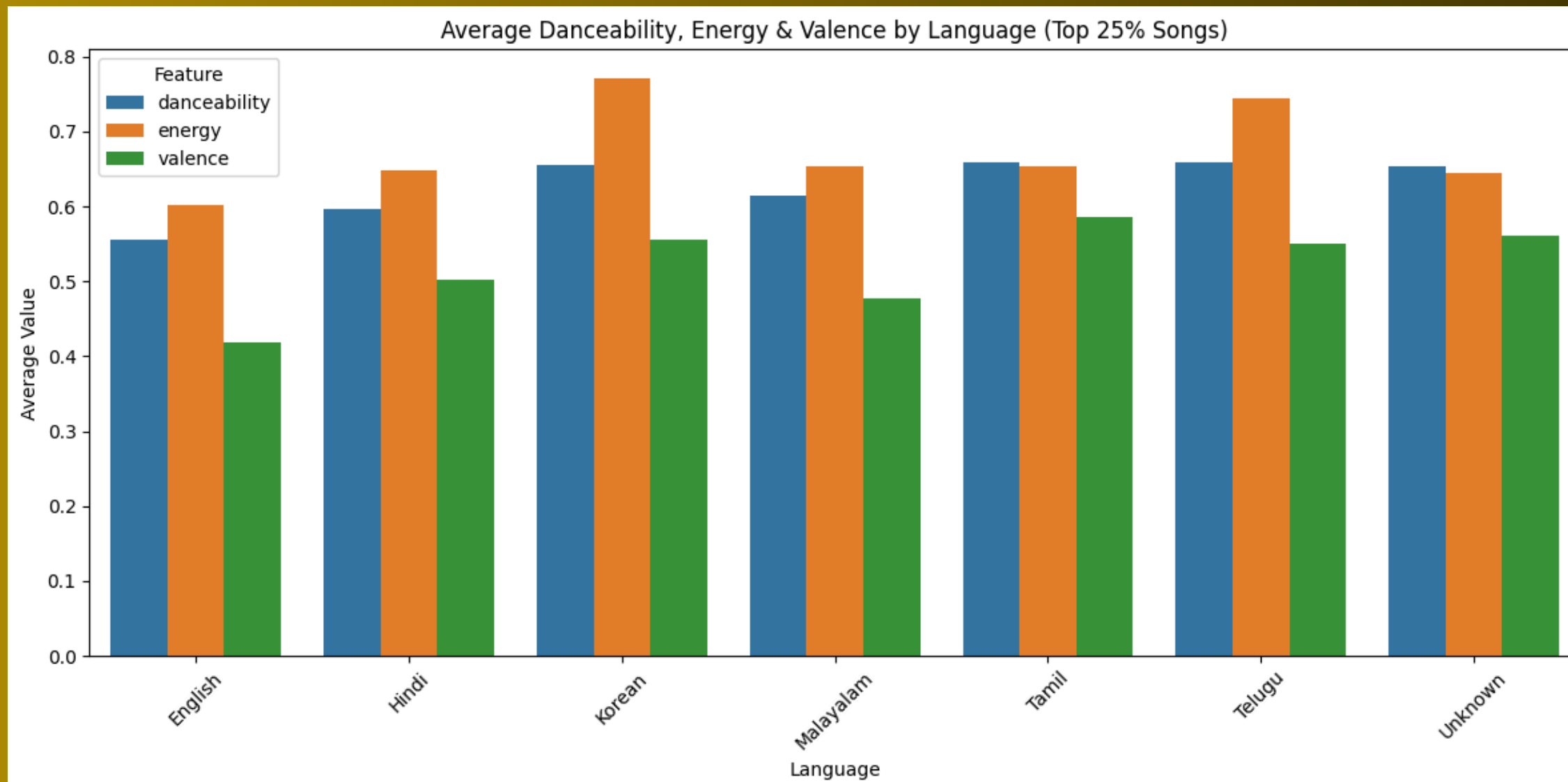


Based on the "Pairwise Relationships of Key Audio Features" plot, the main insight is the strong, linear positive relationship between **Valence (mood/positiveness) and Energy.**

Key observations:

- **Valence vs. Energy (third row, second column):** There is a clear, positive correlation, meaning songs with higher energy levels tend to have a more positive mood.
- **Danceability vs. Energy (second row, first column):** These two features also show a clear positive correlation, indicating that more energetic songs are also generally more danceable.
- **Acousticness** seems to have a weak or non-linear relationship with most other features, though it shows a slight tendency toward higher popularity when lower (first column of the fourth row).
- **Popularity** is heavily concentrated at lower scores, as seen in its own distribution plot (bottom right).

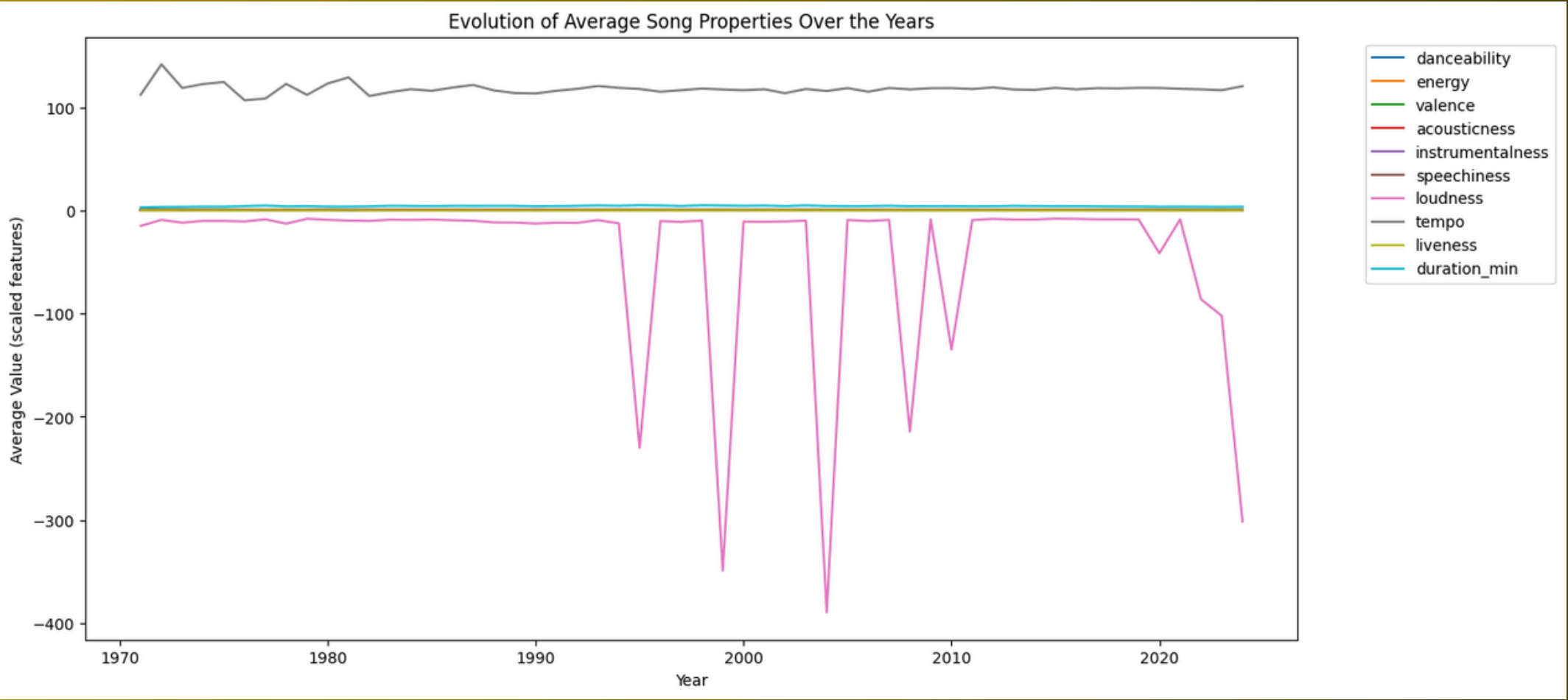
Comparing Musical Vibes Across Languages



Based on the "Average Danceability, Energy & Valence by Language (Top 25% Songs)" bar chart, the key insight is that **Korean language songs have the highest average Energy and Danceability among the top 25% most popular tracks.**

Key observations:

- **Korean has the highest average Energy (nearly 0.8) and the highest average Danceability (nearly 0.7).**
- **Across almost all languages, Energy is the highest-rated feature, followed by Danceability, with Valence (mood) consistently being the lowest.**
- **Telugu songs also show a high average Energy (nearly 0.75).**
- **English songs have the lowest average Valence (mood), suggesting their most popular tracks are slightly less positive compared to other languages.**

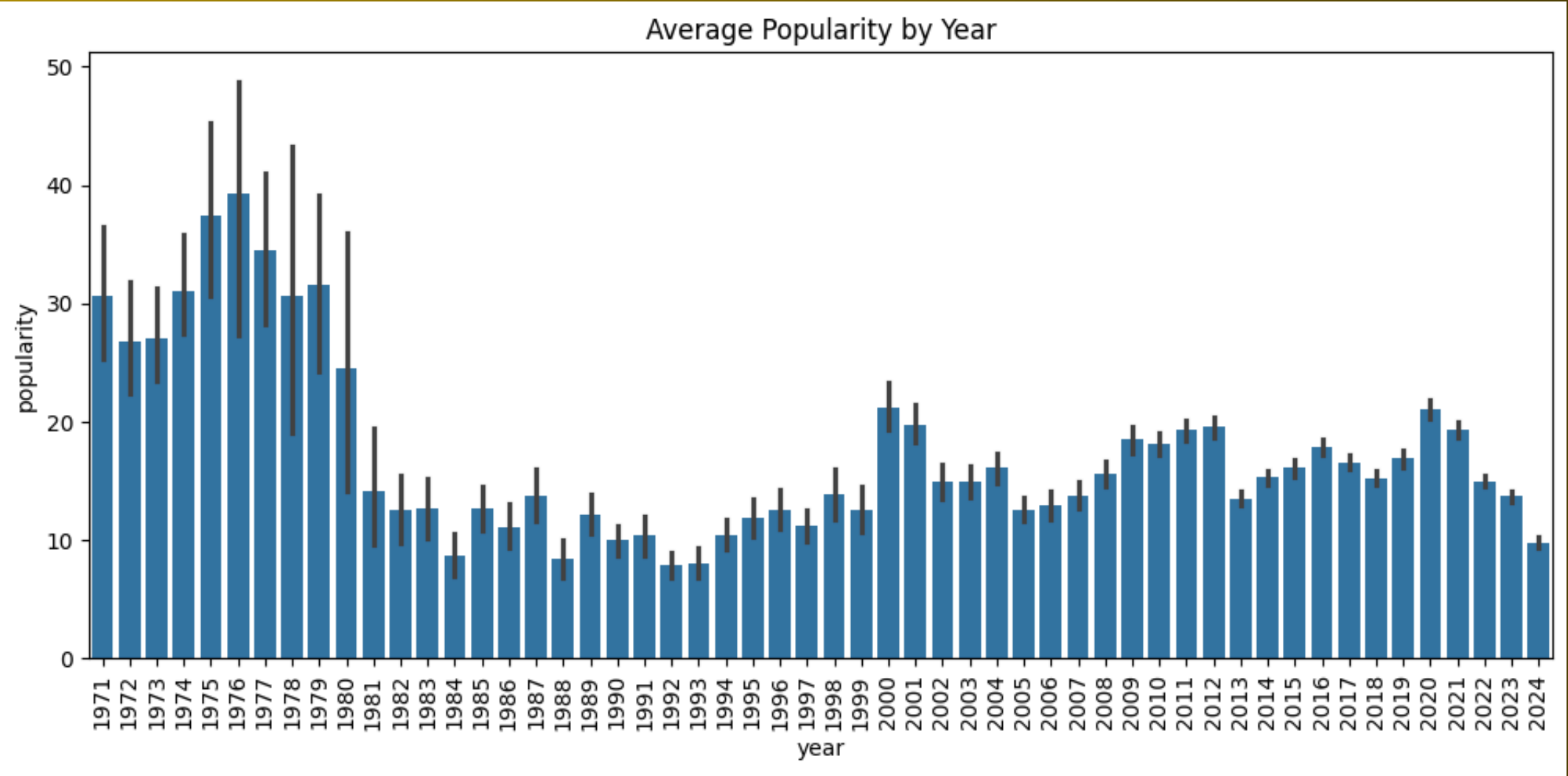


Evolution of Song Features Over Time

Based on the "Evolution of Average Song Properties Over the Years" plot, the primary insight is that most average song properties have remained remarkably stable over the decades, with one major exception.

Key observations:

- **Stability:** Features like tempo (gray line), danceability, energy, valence, and loudness (clustered near 0 or 120 on the scaled axis) show very little year-to-year change from 1970 to 2024.
- **Volatile Feature:** The average value for instrumentalness (purple/pink line) is highly erratic, showing dramatic, year-on-year swings, especially from the late 1990s onward, with large, sudden drops (suggesting occasional years with a very low average instrumentalness score)



Evolution of Music Popularity Over Time

Based on the "Average Popularity by Year" bar chart, the primary insight is the dramatic and sustained decline in the average popularity of songs released since 1980, compared to the 1970s.

Key observations:

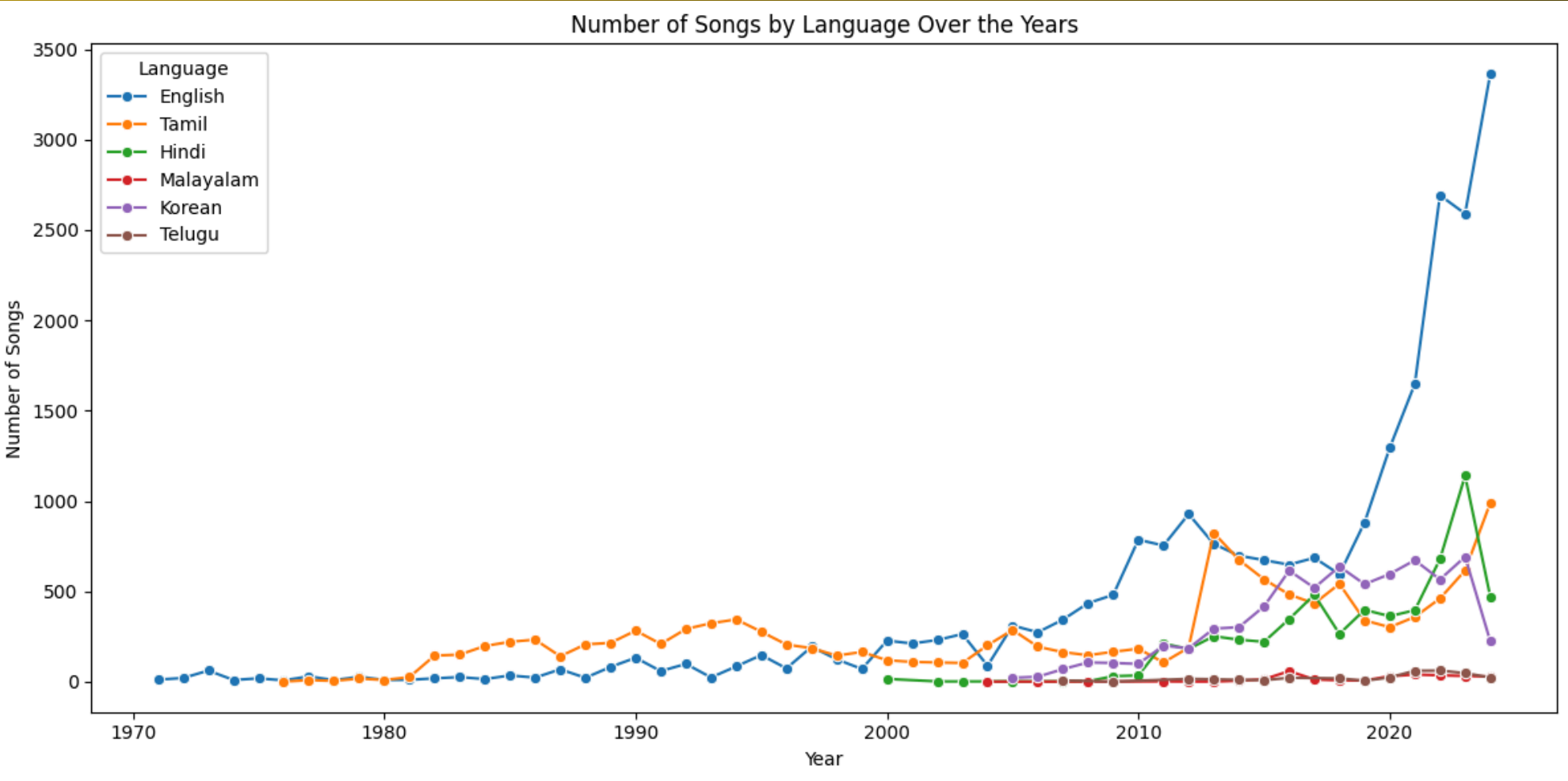
- **Peak Popularity:** The highest average popularity scores (approaching 50) occurred in the mid-1970s (specifically around 1976-1977).
- **Initial Drop:** There was a sharp drop in average popularity from 1980 onwards.
- **Modern Trend:** Since the early 1990s, the average popularity has generally stabilized and gradually increased, but it has never consistently returned to the high levels seen in the 1970s.

Growth of Songs by Language Over the Years

Based on the "Number of Songs by Language Over the Years" line chart, the most compelling insight is the explosive, recent growth in English songs, which has dramatically pulled away from all other languages.

Key observations:

- **Dominant Recent Growth:** English songs experienced a massive surge starting around 2017, rising from under 1,000 to over 3,000 by 2023, far exceeding every other language.
- **Historical Leadership:** Tamil had the highest number of songs for much of the period from the mid-1980s to the early 2010s.
- **Rise of Others:** Since around 2015, Hindi and Korean have also seen significant increases, but neither approaches the current volume of English.

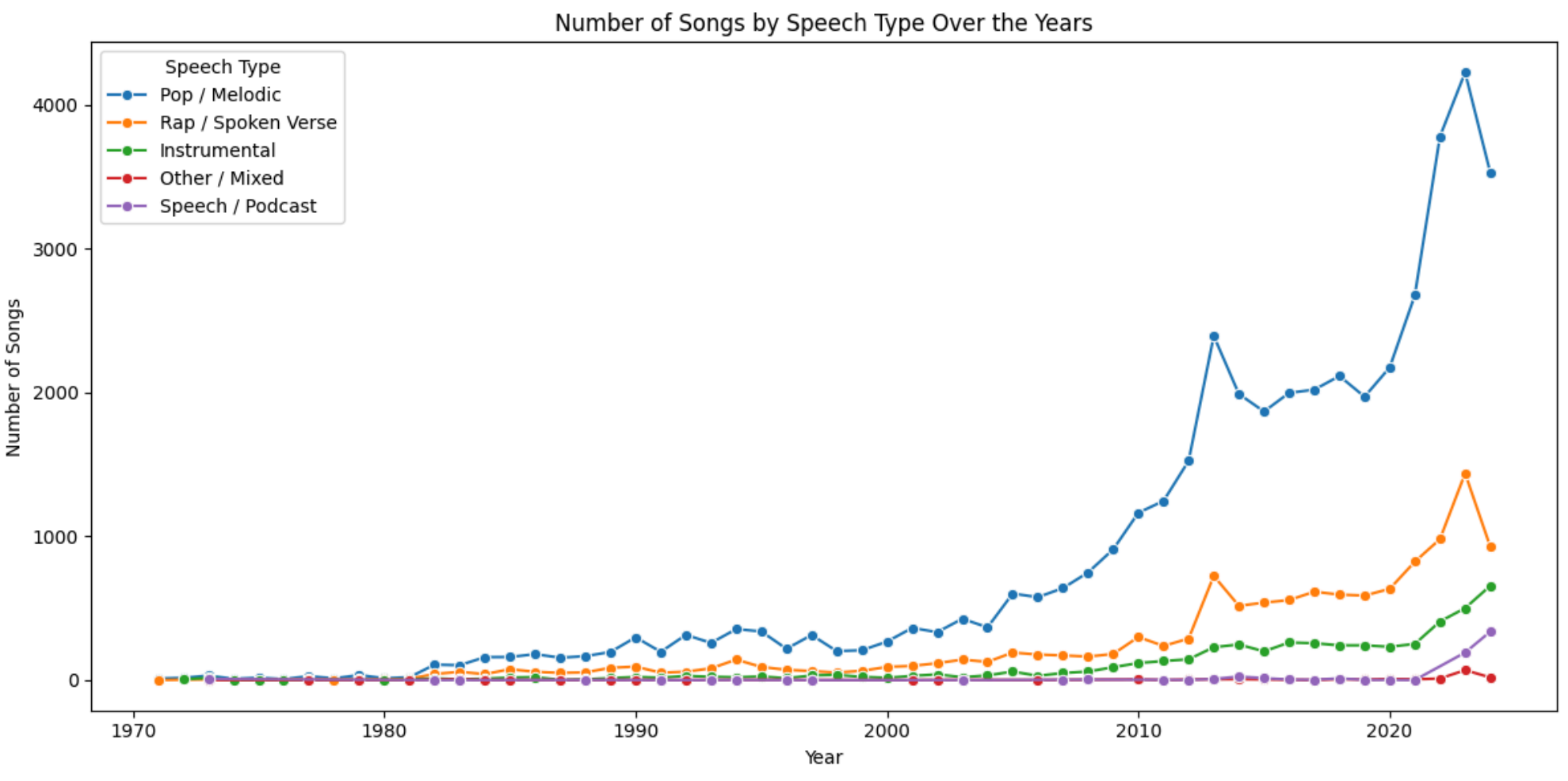


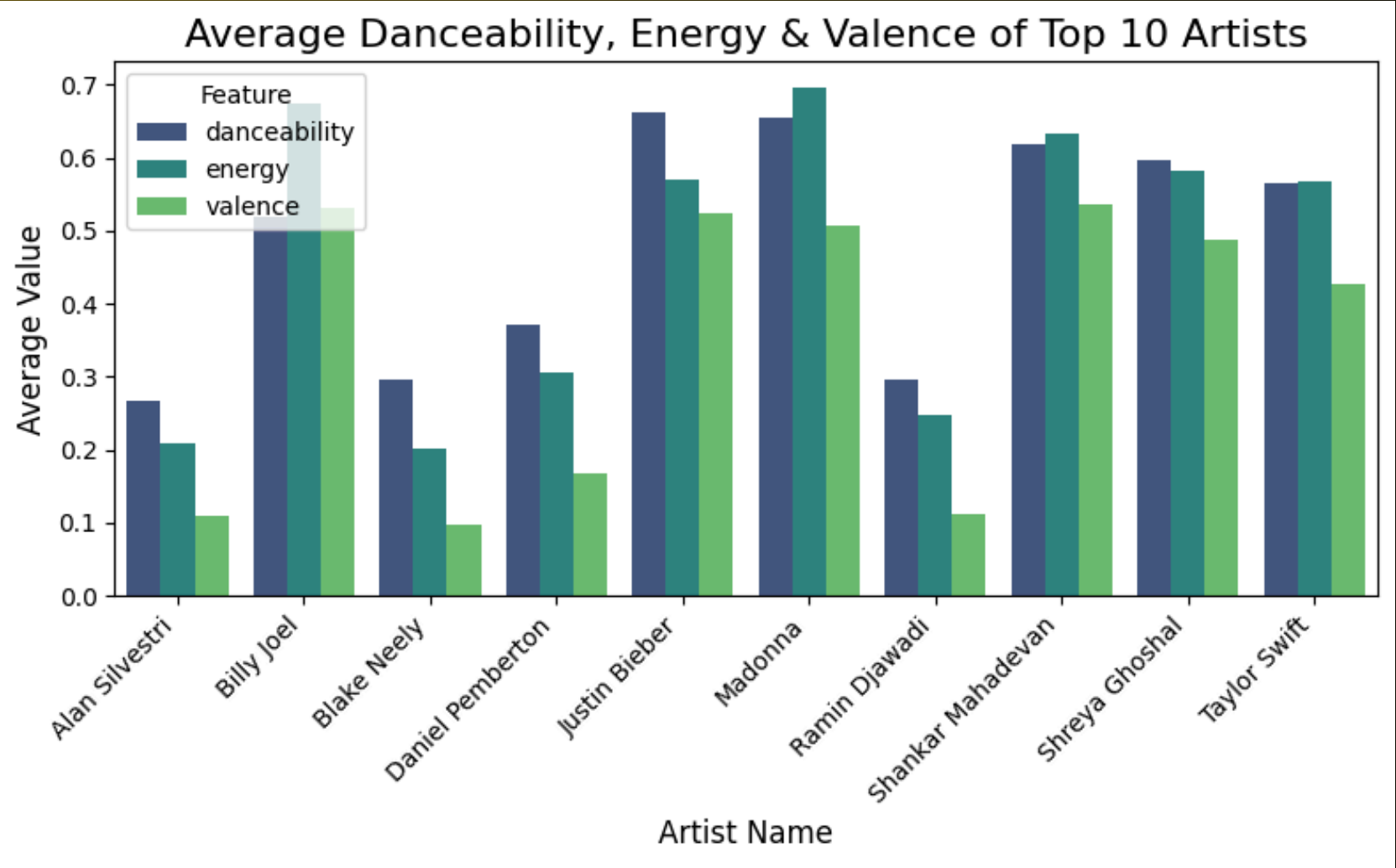
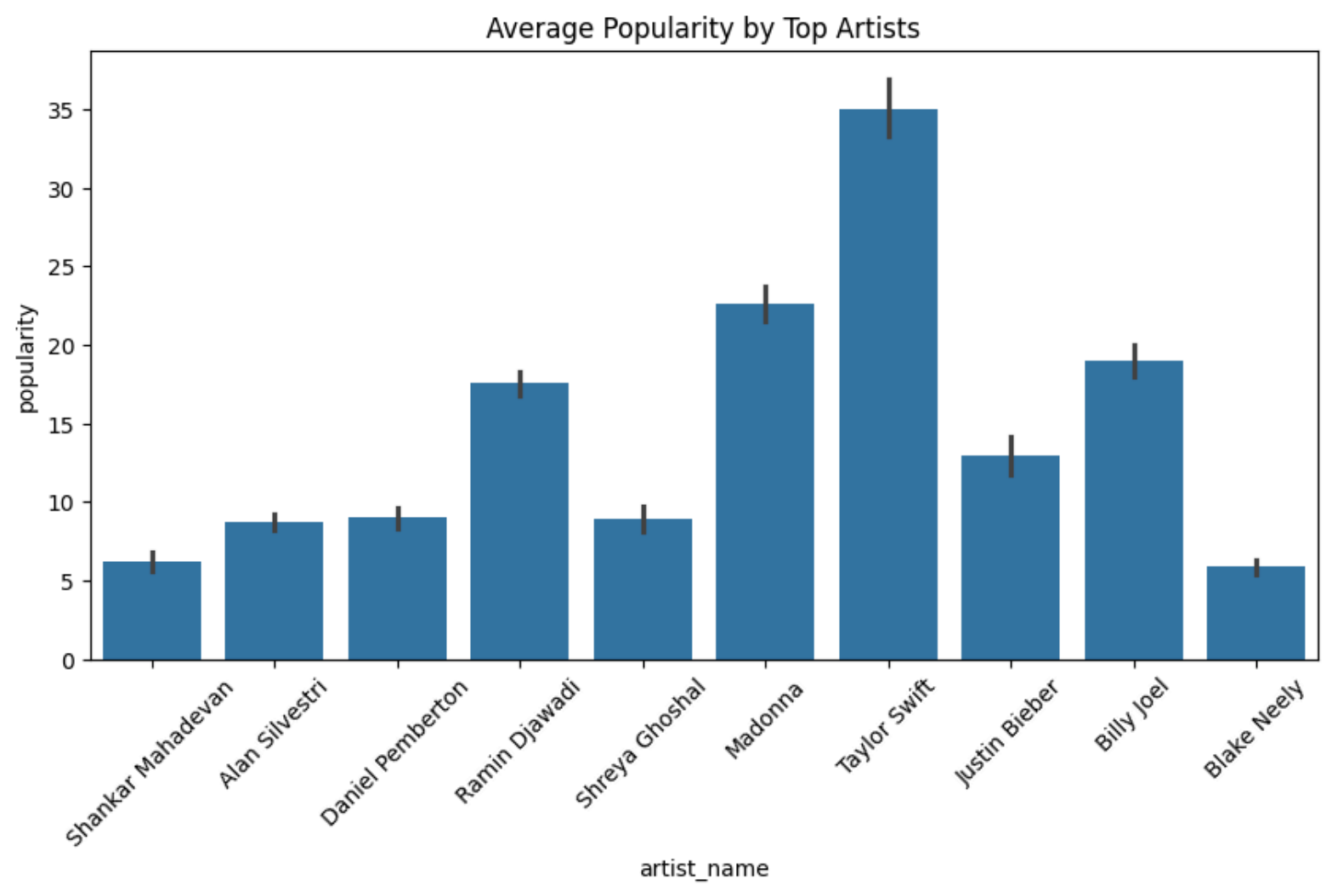
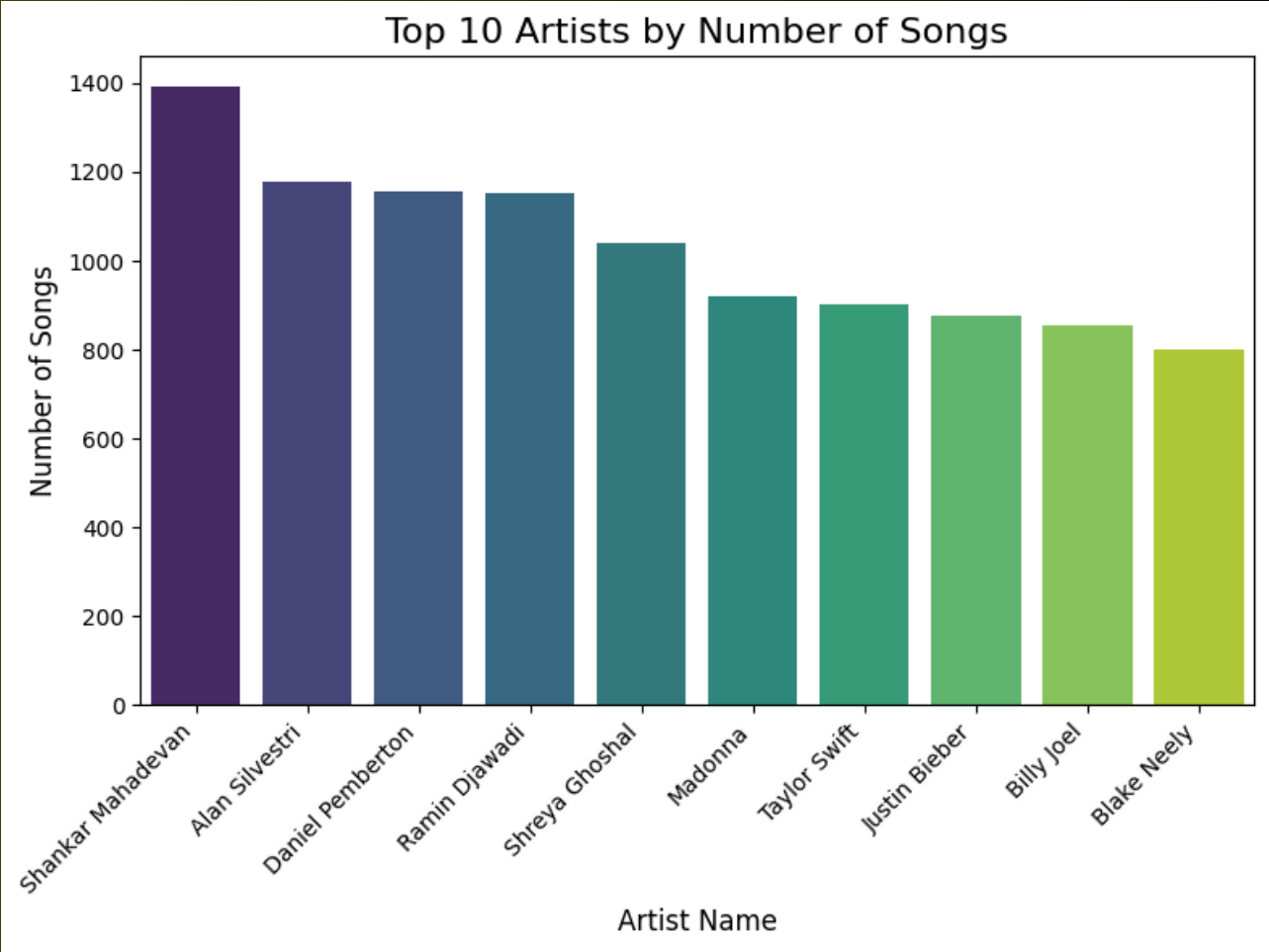
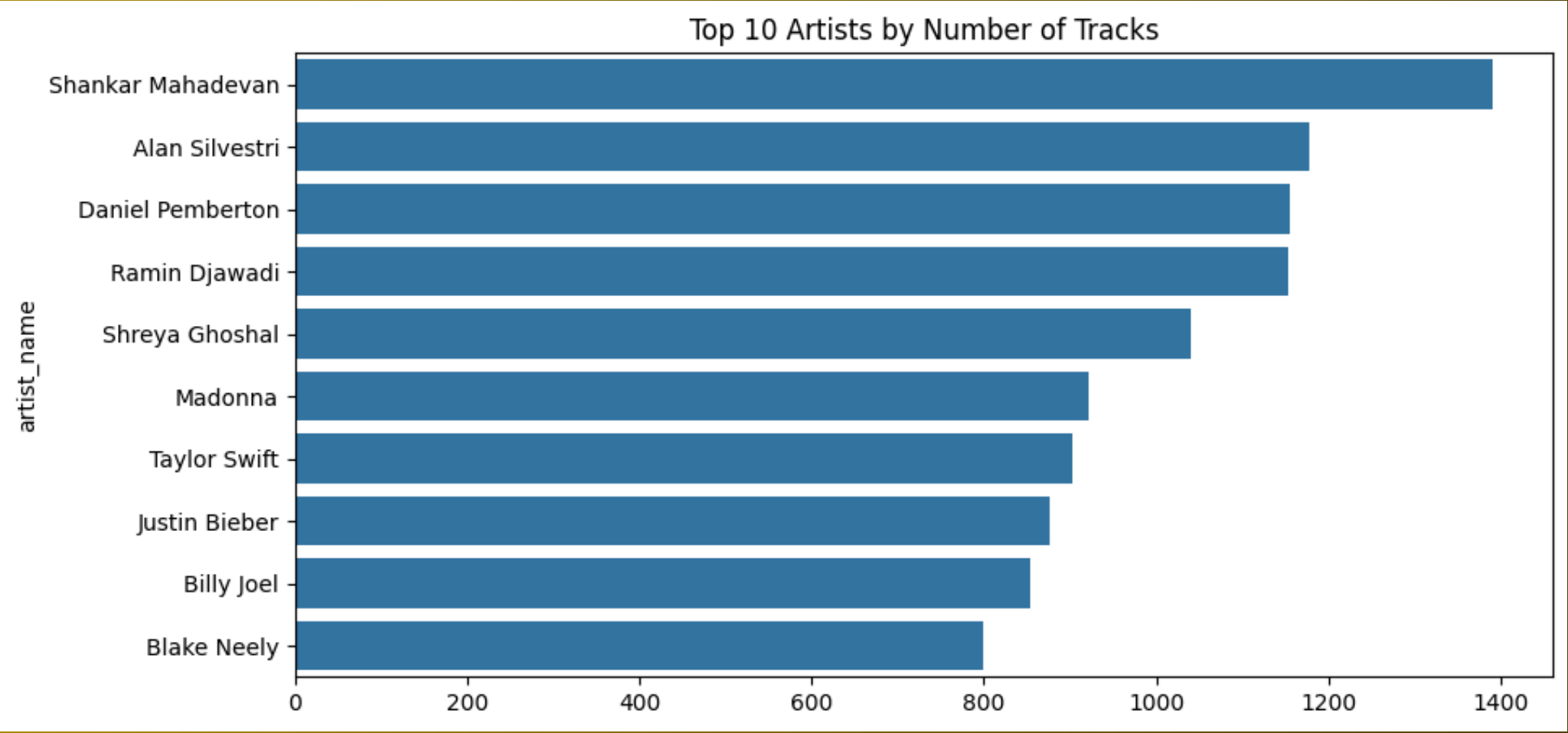
Yearly Distribution of Speechiness in Music

Based on the "Number of Songs by Speech Type Over the Years" line chart, the most significant insight is the massive, accelerating growth in the number of "Pop / Melodic" songs, especially since 2015.

Key observations:

- **Dominant Growth:** The Pop / Melodic category far outpaces all others, skyrocketing from around 2,000 songs in 2015 to over 4,000 by 2023.
- **Secondary Growth:** The Rap / Spoken Verse category also shows substantial growth, becoming the second most numerous type, particularly after 2015.
- **Overall Trend:** The total number of songs recorded and categorized (in all major speech types) has been on a generally increasing trend since the mid-1980s, indicating a significant expansion of the music catalog over time.





Thank You

I hope you enjoyed my analysis of the Spotify dataset.

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